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CS23532-COMPUTER NETWORKS-LAB MANUAL

PRACTICAL - 8

AIM:

To use Nmap (Network Mapper) on the TryHackMe platform to discover live hosts in a network using different host-discovery techniques such as ARP, ICMP, TCP, and UDP scans, and to analyze the scan results.

PROCEDURE:

Step 1: Open TryHackMe

1. Log in to the **TryHackMe** platform.
2. Open the relevant Nmap/network-scanning room.
3. Start the target machine.
4. Start the **AttackBox** or connect to the TryHackMe network through VPN.
5. Open the terminal.

Step 2: Check Nmap installation

Execute: nmap –version

Step 3: Identify the target

Use the target IP address provided by the TryHackMe machine.

Step 4: ARP Host Discovery

ARP discovery is mainly useful when scanning hosts on the same local Ethernet network.

Commands:

i) sudo nmap -sn -PR 10.48.163.95

ii) **For a subnet:** sudo nmap -sn -PR 10.48.163.0/24

Step 5: ICMP Host Discovery

ICMP Echo Requests can be used to determine whether a host is reachable.

Commands:

i) sudo nmap -sn -PE 10.48.163.95
ii) sudo nmap -sn -PE 10.48.163.0/24

Result:

Host is up

Step 6: TCP Host Discovery

TCP-based discovery can be used when ICMP requests are blocked.

Commands:

i) TCP SYN Discovery: sudo nmap -sn -PS80,443 10.48.163.95

Result:

Host is up

Step 7: TCP ACK Discovery

Another TCP-based discovery method uses TCP ACK packets.

Commands:

i) sudo nmap -sn -PA80,443 10.48.163.95

Result:

Host is up

Step 8: UDP Host Discovery

UDP probes can also be used for host discovery.

Commands:

i) sudo nmap -sn -PU53,161 10.48.163.95

Result:

Host is up.

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

Step 9: Combined Host Discovery

Nmap can also use multiple discovery techniques.

Commands:

i) sudo nmap -sn -PE -PS80,443 -PU53 10.48.163.95

This performs host discovery using:

- ICMP Echo
- TCP SYN
- UDP probes

Result:

Host is up (0.10s latency).

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

Step 10: Basic Nmap Scan for Further Verification

A normal Nmap scan can be performed to identify open TCP ports.

Commands:

i) nmap 10.48.163.95

Result:

Host is up (0.10s latency).

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

443/tcp open https

Nmap done: 1 IP address (1 host up) scanned in 2.15 seconds

Result:

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Thus, Nmap was successfully used on the TryHackMe platform to discover live hosts using ARP, ICMP, TCP, and UDP host-discovery techniques. The scan results were analyzed to determine the availability of the target hosts and, where applicable, the open ports and services running on them.